

IN615008:
Switching, Routing and Wireless Essentials
Assignment (40%)

Student Name:

Due Date/Time: Wednesday 3 June

A 10% penalty will be applied for each day late.

Course credits: 15
Level: 6

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Instructions

You are required to complete this assignment and submit a digital copy of this document, along with the Packet Tracer file of the completed network configuration. Your digital copy of this document should include the answers to all questions as indicated. The two files will need to be emailed to mholtz@op.ac.nz by the date/ time listed.

Learning Objectives

1. Design and build a network for availability and redundancy.
2. Configure and troubleshoot flexible local area networks using virtual and wireless technologies.

Background

Choam Corporation is redesigning its network to take advantage of a hierarchical network design and the performance and scalability benefits of OSPF. You have been requested to carry out the redesign and ensure it meets all specified requirements.

The physical topology has already been decided upon. You are required to make decisions about the logical topology and create the resulting network within Packet Tracer for testing purposes. Choam Corporation has also requested that you answer some questions pertaining to the design, the technologies employed and possible future direction.

Requirements

The network is to be comprised of a Head Office (HO) LAN that is connected to the internet and one additional remote WAN site. The HO LAN utilises a collapsed core architecture which combines both distribution and core layers.

HO Requirements

- 2 user VLANs (Sales, Admin). Each user VLAN will have a maximum size of 20 usable IP addresses. Ensure these networks can be summarised in as small a contiguous block as possible.
- 1 server VLAN. The server VLAN will require a maximum of 13 usable IP addresses.
- Use private IP addressing from 10.8.0.0/16 unless otherwise specified.
- The company wishes to conserve IP Addresses where possible for future use. You must use VLSM for IP addressing.
- All HO switches require a management address within a dedicated management VLAN. The management network is a /28 IP network.
- OSPF will run between the WAN edge routers and between HO and Site2.
- Access and Distribution switches are layer 2 (2960) switches.
- You are required to decide upon a naming convention, for the network devices and the end hosts, that is logical and consistent.
- 2 servers in a dedicated VLAN will each be connected to a different access switch.
- For the purpose of the packet tracer proof of concept, only 1 PC per user VLAN is required per access switch.
- Further requirements are outlined in each section below.

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Site-2 Requirements

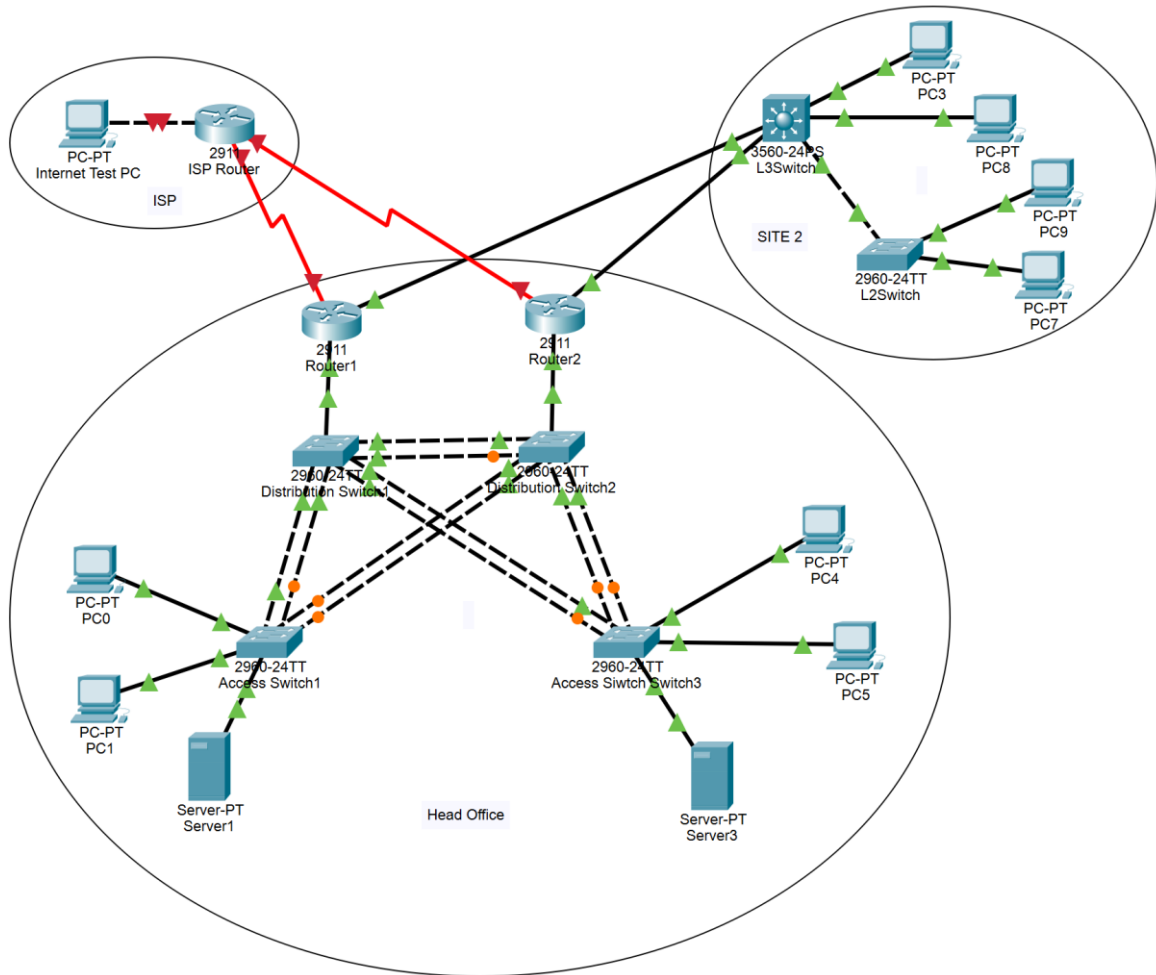
- Use private addressing from 10.9.0.0/24 unless otherwise specified.
- 6 employees in the Site2 Admin department.
- 10 employees in the Site2 Sales department.
- No management VLAN is required.
- You must use VLSM for IP address allocation.
- One switch is Layer 3 (3560), the other is Layer 2 (2960).
- You are required to decide upon a naming convention, for the network devices and the end hosts, that is logical and consistent.
- There are no servers located in Site2.
- 2 PCs are cabled per switch. Each Switch will support 2 user VLANs.
- The link between the 2 switches is a layer 2 trunk.
- Further requirements are outlined in each section below.

IP address assignment

- All PCs will be dynamically assigned IP addresses.
- All other IP addresses will be statically assigned.

Network Topology

The physical topology is shown here. Link status indicators are not accurate as the devices shown are not configured.



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Section One: Document Logical Design [21 marks]

1. You are required to produce documentation (use tables, See the Appendix for example templates.) for the following:
 - All device names
 - Interface addressing (all statically assigned addresses of routers, switches and servers)
 - VLAN interface assignment [6 marks]
2. You are required to produce documentation (use tables, See the Appendix for example templates.) for the following:
 - All IP Networks (include Network IP, Default Gateway, Broadcast, Max usable hosts, mask, purpose). Note you will need to perform subnetting as required to create the required number of networks outlined in the above specifications. [15 marks]

Section Two: Head Office LAN [63 marks]

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1. Cable 2 access layer switches and 2 distribution switches (all are 2960 Layer switches). Ensure the access switches are cabled to each distribution switch. Ensure the distribution switches are cabled to each other. Ensure an adequate number of cables are used to facilitate etherchannel in step 9. [5 marks]
2. Ensure the switches are correctly configured with a hostname, management address and default gateway. [5 marks]
3. Configure the LAN for Rapid-PVST+. Configure the 2 User VLANs, Server VLAN and a Management VLAN. Choose and configure a switch to be the root bridge for the 2 User VLANs and the second switch to be root for the remaining VLANs. Choose and configure secondary root bridges for each VLAN. [15 marks]
4. Describe how the root bridge, designated ports and root ports would be determined by default. Provide captured output to confirm the root bridge in your topology. [10 marks]
5. Cable and configure at least 1 PC per User VLAN on each of the access switches. The PC IP addressing is dynamic. [2 marks]
6. Cable and configure 2 servers, one in each of the Access switches. Use static IP addressing. [4 marks]
7. Ensure all appropriate ports have port-fast configured. What is the purpose of this command? [5 Marks]
8. Ensure all appropriate ports have BPDU-guard configured. What is the purpose of this command? [5 marks]
9. Implement PAgP and 802.1q trunking between the two distribution switches ensuring at least 2 physical cables are used. Implement LACP and 802.1q trunking between each of the access switches and distribution switches (2 physical links per channel). Ensure the native VLAN is non default. Ensure only user, server and management VLANs are permitted on each trunk. [10 marks]
10. In your documentation provide captured output showing which ports have been aggregated and the type of aggregation (for all port channels). [2 marks]

Section Three: Head Office Routing [50 marks]

1. Add 2 head office routers. Each router is cabled to a different distribution switch. Ensure the HO routers are configured with a hostname, logging synchronous, no ip domain lookup & SSH access (including admin user). [5 marks]
2. Configure each Router for inter VLAN routing using ROAST. [10 marks]
3. Configure HSRP for each VLAN. Ensure that the active gateway is connected to the root bridge for that VLAN. [10 marks]
4. Ensure pre-emption is supported. Document what this does. [5 marks]
5. Show capture output that displays the standby status on both HO routers. [4 marks]

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6. Configure OSPF AREA 0 on each router for the HO VLAN networks. Ensure Router-IDs are configured. Show output evidence confirming the OSPF adjacencies are operational between the two HO routers. [10 marks]
7. Research the function of DR & BDR roles within OSPF. Explain why they exist and the default process for DR election. [6 marks]

Section Four: Site2 LAN [20 marks]

1. Cable 2 PCs to each of the Site2 switches (3560 & 2960). Connect the 2 switches together using 1 ethernet cable. Ensure each Site 2 switch has a Hostname [2 marks]
2. Configure an Admin and Sales VLAN on each switch. Configure 1 PC in each VLAN on each switch, ensure IP addressing is dynamic. [4 marks]
3. Configure inter VLAN routing on the Site2 L3 switch. Ensure the L3 switch supports routing.[5 marks]
4. Configure the link between the 2 switches as a trunk that only permits the Admin and Sales VLANs . [4 marks]
5. Configure switchport security for all Site2 switchports connected to PCs. Each port is to allow only 2 dynamically learned MAC addresses. These are to be auto populated into the running-config. Ensure there is a notification if a violation occurs. [5 marks]

Section Five: WAN [28 marks]

1. Cable and configure two ethernet links (WAN) between HO LAN and the Site2 L3 switch as per the diagram. Ensure the Site 2 Switch WAN interfaces are routed. Use the following IP networks for the links: 172.16.1.0/30 & 172.16.1.4/30. [2 marks]
2. Configure OSPF area 0 for the WAN links between the HO Routers and the Site2. Ensure router IDs are configured. Show output evidence confirming the OSPF adjacencies are operational. [10 marks]
3. Configure the Site2 L3 switch to advertise the Site 2 VLAN networks via OSPF within Area 2 (the Site L3 Switch will be an ABR). Ensure OSPF packets are not sent over user VLAN interfaces in site2. [6 marks]
4. Configure the Site 2 L3 Switch to advertise a single OSPF summary address representing the Site2 user VLAN networks. Provide evidence of the summary routes. [5 marks]
5. Configure OSPF such that the Head Office Router 2 is the preferred path to and from Site2. Explain how you have done this, do not rely on default behaviour. [5 marks]

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Section Six: DHCP [17 marks]

1. Configure DHCP on HO Router2 connected to Site2 to support the 2 HO User VLANs and the 2 Site2 User VLANs. Ensure to exclude addresses that have already been statically configured such as gateways. [8 marks]
2. Configure the Site2 switch to allow Site2 PCs to obtain IP addresses from the HO router. [3 marks]
3. Document output from the HO Router2 confirming DHCP leases for HO and Site2. [3 marks]
4. Explain the difference between stateless and stateful DHCPv6. [3 marks]

Section Seven: Internet Routing [15 marks]

1. Cable ISP router to HO Router1 and Router 2 using serial. Configure the ISP router with a hostname. Configure the links between the HO routers and the ISP router with a clockrate and the 192.168.1.0/30 & 192.168.1.4/30 networks, ensure the ISP end uses the first available IP. [3 marks]
2. Configure a static default route on HO Router1 & Router 2 pointing to the IP address assigned to the ISP router end of the respective links. Advertise the statics using OSPF on each router. [4 marks]
3. Configure a total of 4 static routes on the ISP router.
A summary IP network address representing the HO LAN networks pointing to the outbound interface facing HO Router 1.
A floating static backup summary route for the HO LAN networks pointing to the outbound interface facing HO Router 2.
A summary IP network address representing the Site 2 LAN networks pointing to the outbound interface facing HO Router 2.
A floating static backup summary route for the Site 2 LAN networks pointing to the outbound interface facing HO Router 1.
pointing to the outbound interface facing HO. [6 marks]
4. Configure the internet test host directly connected to the ISP router. The host uses the IP address of 209.100.100.3/29, the ISP router interface connected to the PC uses 209.100.100.1/29. [2 marks]

Section Eight: IPv6 [20 marks]

1. IPv6 Dual stack. You have been tasked with adding IPv6 to the Admin and Sales networks in the Head Office Site. You are to subnet the 2001:DB8:100::/48 network as required for all addressing needs. HO Router 1 is to be configured as an IPv6 DHCP server that will provide dynamic addressing for PCs in the HO User VLANs. Router 2 is not required to run IPv6 (HSRP is not required).

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2. The ISP router and Internet test PC are to be configured with IPv6 addressing. Static routing is to be configured as required to ensure that HO User VLAN PCs can connect to the Internet Test PC.
3. Provide evidence of IPv6 connectivity between a PC in the HO Admin VLAN and the HO Sales VLAN. [10 marks].
4. Provide evidence of IPv6 connectivity between HO user PCs and the Internet Test PC. [10 marks]

Section Nine: Testing / Final Connectivity [8 marks]

1. Test that you can ping from a PC within each Head Office and Site2 VLAN to a PC in each other HO and Site2 VLAN. Test that a PC in each user VLAN can ping a Server. Test that HO LAN PCs and Servers can ping the internet PC. If any pings fail, then troubleshoot as necessary to resolve the issues. Provide a table of your results. If any pings are still failing, then describe what is happening and why you think it is happening. [8 marks]

Section Ten: Formatting [8 marks]

1. Ensure your packet tracer network diagram is commented and supporting documentation is formatted neatly and comprehensible. [8 marks]

Assignment of Marks

You will be assigned marks as follows:

- **Total Marks available** 250 marks

Section	Topic	Mark
Section One	Logical Design	21
Section Two	HO LAN	63
Section Three	HO Routing	50
Section Four	Site2 LAN	20
Section Five	WAN	28
Section Six	DHCP	17
Section Seven	Internet Routing	15
Section Eight	IPv6	20
Section Nine	Testing	8
Section Ten	Formatting	8

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Appendix – Documentation Templates

Add rows as required.

Network Summary (all IP subnets)

Network Name	VLAN	Number of host addresses required	Network Address	Subnet Mask	Max Number of Hosts Possible	Assignable Range	Gateway Address

HQ Router1 Name:

Interface/Sub Interface	Network Name	Description and Purpose	Network Address	Interface IP Address	Subnet Mask

HQ Router2 Name:

Interface/Sub Interface	Network Name	Description and Purpose	Network Address	Interface IP Address	Subnet Mask

Site2 Switch Name:

Interface	Network Name	Description and Purpose	Network Address	Interface IP Address	Subnet Mask

ISP Router Name:

Interface	Network Name	Description and Purpose	Network Address	Interface IP Address	Subnet Mask

HQ Distribution Switch1 Name:

Switch Mgmt IP address:

Management VLAN:

Port/Number	Description and Purpose	Trunk / Access	VLAN / VLANs allowed

HQ Distribution Switch2 Name:

Switch Mgmt IP address:

Management VLAN:

Port/Number	Description and Purpose	Trunk / Access	VLAN / VLANs allowed

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HQ Access Switch1 Name:

Switch IP address:

Management VLAN:

Port/Number	Description and Purpose	Trunk / Access	VLAN / VLANs allowed

HQ Access Switch2 Name:

Switch IP address:

Management VLAN:

Port/Number	Description and Purpose	Trunk / Access	VLAN / VLANs allowed

HQ Access Switch3

Switch IP address:

Management VLAN:

Port/Number	Description and Purpose	Trunk / Access	VLAN / VLANs allowed

Site2 MLS Switch Name:

Port/Number	Name / Description and Purpose	Interface IP Address	Subnet Mask	Trunk / Access	VLAN / VLANs allowed

Static Server Addresses

Server Name	IP address	Subnet Mask	Gateway